

Bacillary Angiomatosis (BA)

Currently, the prevalence of bacillary angiomatosis (BA) in North America and Western Europe is very low, primarily due to improved immune function in HIV-infected patients on highly active antiretroviral therapy (HAART) and the use of prophylactic treatments against organisms such as *Mycobacterium avium complex* (MAC).

Clinically, the cutaneous lesions of BA are red to violaceous, dome-shaped papules, nodules, or plaques that resemble Kaposi sarcoma or pyogenic granuloma. These lesions range in size from a few millimeters to 2–3 cm in diameter and represent dermal vascular proliferations with overlying thinned or eroded epidermis. Less commonly, dome-shaped subcutaneous masses may occur without the characteristic red color seen in more superficial vascular lesions.

Lesions are typically soft to firm and may be tender on palpation. The number of lesions varies widely—from a single lesion to more than 100, and in rare cases, exceeding 1000. Nearly any cutaneous site can be involved, although the palms, soles, and oral cavity are usually spared.

Following hematogenous or lymphatic dissemination, *Bartonella henselae* and *Bartonella quintana* can cause systemic disease, including soft-tissue masses, bone marrow involvement, lymphadenopathy, splenomegaly, and hepatomegaly. Internal organ involvement may occur with or without concurrent cutaneous lesions.

The clinical course of BA is variable. In some patients, lesions regress spontaneously. However, as with other opportunistic infections in HIV, BA may recur, especially in the setting of persistent immunosuppression.

Treatment

The antibiotics of choice for BA are: - **Erythromycin**: 250–500 mg orally four times daily - **Doxycycline**: 100 mg orally twice daily

Treatment should continue until complete resolution of lesions, which typically takes 3 to 4 weeks. Alternative effective agents include rifampin, ciprofloxacin, and trimethoprim-sulfamethoxazole (TMP-SMX), particularly for patients intolerant to first-line therapies.

Secondary prophylaxis is recommended for patients with recurrent BA.

Fungal Infections

Cutaneous fungal infections in HIV disease manifest as: - Superficial infections (dermatomycoses) - Invasive cutaneous infections - Systemic fungal infections with hematogenous spread to the skin

Superficial pathogens such as *Trichosporon beigelii* can disseminate as immune function declines. Dissemination may present atypically; for example, *Trichophyton rubrum* has been reported to disseminate and clinically mimic Kaposi sarcoma.

The widespread use of HAART has significantly reduced the incidence of these infections. Additionally, drug interactions between HAART and antifungal agents appear to be less clinically significant than previously believed.

Dermatophytoses

Trichophyton rubrum is the most common cause of **proximal subungual onychomycosis**, an infection involving the undersurface of the proximal nail plate. This condition is strongly associated with HIV infection and should prompt consideration of HIV testing.

In advanced HIV disease, epidermal dermatophytosis can be extensive, persistent, and recurrent. Without restoration of immune function, these infections tend to be chronic.

Mucocutaneous Candidiasis

Cutaneous *Candida* infections such as intertrigo are relatively uncommon in HIV-infected adults, though the prevalence may increase in those with HAART-associated diabetes mellitus.

Candida infections frequently affect moist, keratinized areas such as the anogenital region. **Angular cheilitis**, characterized by inflammation at the corners of the mouth, may occur unilaterally or bilaterally. It can appear in isolation or in conjunction with oropharyngeal or esophageal candidiasis.

In HIV-infected children, common presentations include: - Diaper-area candidiasis - Intertrigo in the axillae and neck folds - Chronic *Candida* paronychia of the fingernails, often leading to secondary nail dystrophy (onychia)

Oropharyngeal Candidiasis

Candida colonization of the oropharynx is common in HIV-infected individuals, often occurring in the absence of symptoms. Its presence may warrant HIV testing.

- *C. albicans* is the most frequently isolated species.
- Colonization with non-*albicans* species alone is associated with advanced HIV disease and very low CD4+ T-cell counts.

Oropharyngeal candidiasis serves as a clinical marker of HIV disease progression.